

UNIVERSITY OF ADEN
Department of Electronics and Communication
Even Semester Examinations Academic Year 2021/2022
Subject: Electronics Circuit

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Date: 25/8/2022
Time: 2 Hours

Class: B2CS/E
ANSWER ANY FOUR QUESTIONS

Question 1

Determine if the diode in fig. 1 is forward biased or reverse biased.

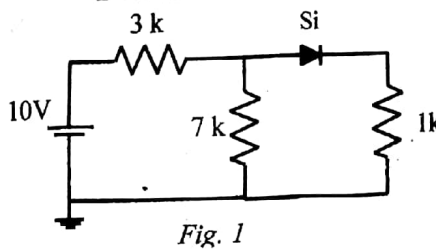


Fig. 1

$$E - IR_1 - IR_2 = 0$$

$$E - I(R_1 + R_2) = 0$$

$$\Rightarrow I = \frac{E}{R_1 + R_2} = \frac{10}{10 \times 10^3} = 1 \times 10^{-3} \text{ A} = 1 \text{ mA}$$

$\text{mA} \rightarrow \text{A}$

Question 2

Sketch and determine the output voltage V_o for the clipping circuit shown in fig 2. (show all Steps of calculation)

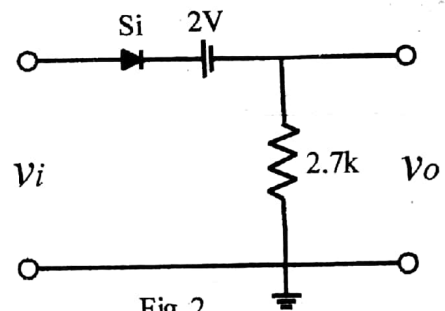
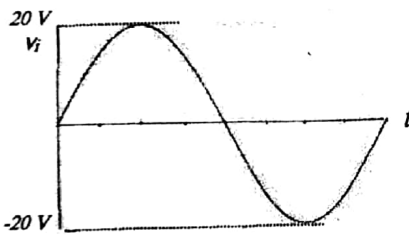


Fig. 2

$$V_{cc} - I_c R_c - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{cc} - I_c R_c - I_E R_E$$

$$18 - 2.92 \times 10^{-3} \times 4.7 \times 10^3 - 2.92 \times 10^{-3} \times 1.2 \times 10^3$$

$$= 14.148 \text{ V}$$

Question 3

For the emitter stabilized bias circuit shown in fig. 3 determine the following values:

$I_B, I_E, V_{CE}, V_C, V_B, V_E$

Side B8

$$V_{cc} - R_B I_B - V_{BE} - I_E R_E = 0$$

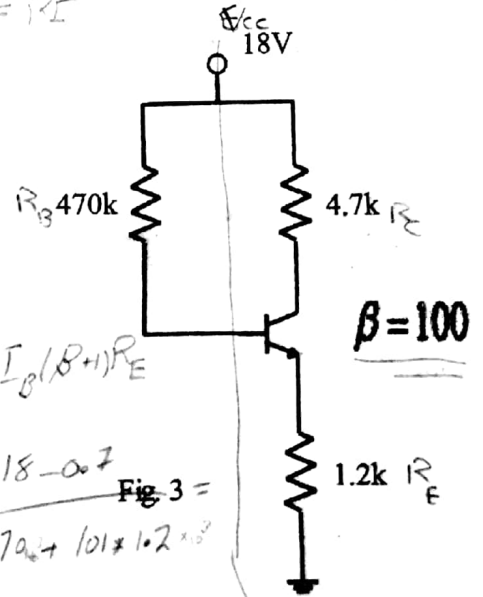
$$V_{cc} - I_B R_B - V_{BE} - I_B (\beta + 1) R_E = 0 \Rightarrow V_{cc} - V_{BE} = I_B R_B + I_B (\beta + 1) R_E$$

$$V_{cc} - V_{BE} = I_B (R_B + (\beta + 1) R_E) \Rightarrow I_B = \frac{V_{cc} - V_{BE}}{R_B + (\beta + 1) R_E} = \frac{18 - 0.7}{470k + 101 \times 1.2 \times 10^3}$$

$$I_E = \beta I_B = 100 \times 2.92 \times 10^{-5} = 2.92 \times 10^{-3} \text{ A}$$

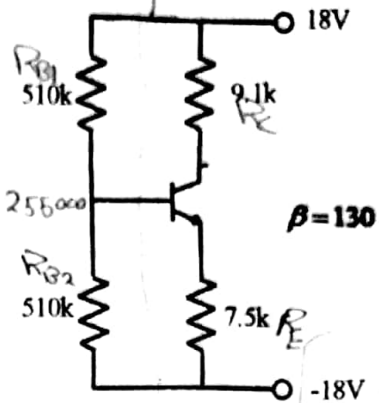
$$V_{CE} = V_{cc} - I_C R_C - I_E R_E = V_{cc} - I_E (R_C + R_E) = 18 - 2.92 \times 10^{-3} (4.7 \times 10^3 + 1.2 \times 10^3) = 0.772 \text{ V}$$

$$V_C = V_{cc} - I_C R_C = 18 - (2.92 \times 10^{-3} \times 4.7 \times 10^3) = 4.276 \text{ V}$$



Question 4

For the common emitter amplifier circuit shown in fig. calculate : I_B , I_C , V_E and V_{CE}



$$V_B = \frac{510 \times 10^3 \times 18}{510 \times 10^3 + 510 \times 10^3}$$

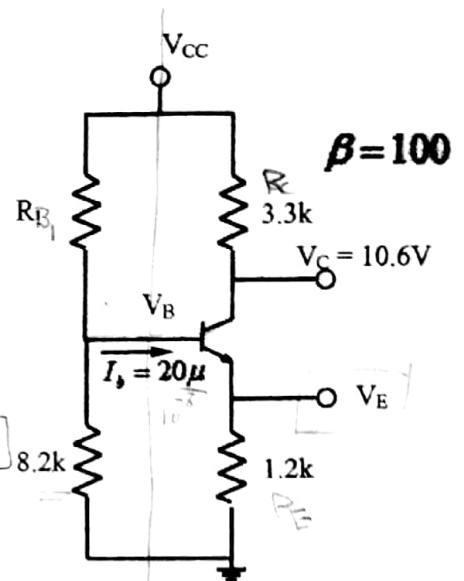
$$R_{th} = \frac{510 \times 10^3 \times 510 \times 10^3}{510 \times 10^3 + 510 \times 10^3} = 255000$$

$$V_B - V_{BE} - I_B R_{th} - I_E R_E - V_2 = 0$$

$$I_B = \frac{V_B - V_{BE} - V_2}{R_{th} + (\beta + 1)R_E}$$

Question 5

For the voltage divider biasing network shown in fig. 4 Determine : I_C , V_E , V_{CE} , V_B and R_1



$$V_B = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$V_B = (R_1 + R_2) I_B$$

$$V_C = V_{BE} = R_B I_B \rightarrow R_B = \frac{V_C}{I_B}$$

$$V_C = V_{CC} - I_C R_C \Rightarrow V_{CC} = V_C + I_C R_C \quad \therefore I_C = \beta I_B$$

$$V_{CC} = 10.6 + 2 \times 10^{-6} \times 3.3 \times 10^3 = 10.602 \text{ V}$$

$$V_E = \frac{I_E}{\beta + 1} R_E = (R_1 + R_2) I_B \times 1.2 \times 10^3 = 101 \times 20 \times 10^{-6} \times 1.2 \times 10^3 = 0.1212 \text{ V}$$

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E = 10.602 - 2 \times 10^{-6} \times 3.3 \times 10^3 - 2.02 \times 10^{-5} \times 1.2 \times 10^3 = 8.138 \text{ V}$$

Question 6

For emitter bias configuration shown in fig 6 . calculate : β , V_{CC} , R_B and I_C

$$V_E = V_B - V_{BE} \Rightarrow V_B = V_E + V_{BE} = 0.1212 + 0.7 = 0.8212 \text{ V}$$

$$V_B = \frac{R_{B2} \times V_{CC}}{R_{B1} + R_{B2}} = V_B \frac{R_{B2}}{R_{B1} + R_{B2}} = R_{B2} \times V_{CC} \Rightarrow R_{B1} = \frac{R_{B2} \times V_{CC}}{V_B} - R_{B2}$$

$$\beta = \frac{I_C}{I_B} =$$

$$V_C = V_{CC} - I_C R_C \quad V_{CC} =$$

$$-27.7$$

$$I_E =$$

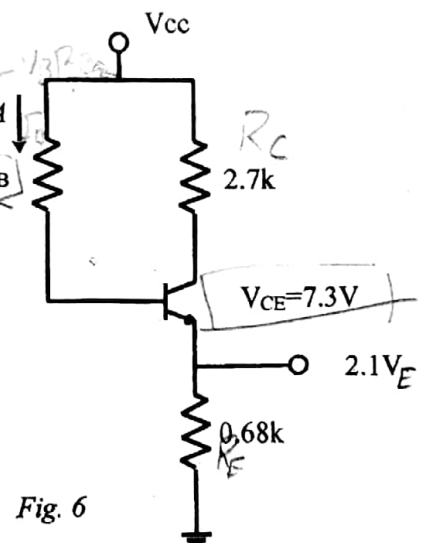


Fig. 6